

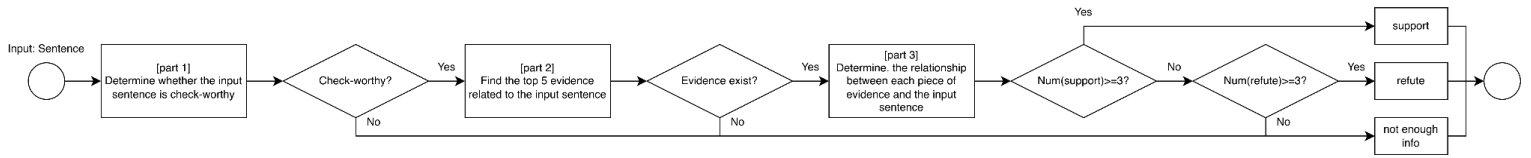
Brief Progress

Member	Progress
Robert Widjaja	- Studied the MNLI, ANLI and SNLI datasets. Label distributions, verifying schemas and cleaning rows with missing premise/hypothesis text - Verified BERT-family tokenization worked on (premise, hypothesis) pairs for cross encoder - Built the unified data loaded which normalized said datasets above into one shared {premise, hypothesis, label} schema (0 = entailment, 1 = neutral, 2 = contradiction) - Built a baseline comparison using TF-IDF + logistic regression (1-2 word n-grams over premise [SEP] hypothesis, plus a hypothesis overlap ratio feature) - Ran baseline results on MNLI-matched dev results shown in another section. - Started working on the cross-encoder wrapper and fine tuning for next steps.
Chloe Lee	Problem with building the knowledge DB - First trial, building the DB locally but it requires a huge amount of memory. As I built this locally, my laptop stopped working so I had to remove everything in the laptop and restart. - Second, building the DB using AWS serverless open search service. I implemented the script for loading the data to open search but it stopped due to a too long connection time. (Took more than 11 hours but did not even load half of them) - For now, I found an open source search library and started testing with this library. If it works, I will use this library instead of building our own search engine. If it doesn't work, maybe we should find another way instead of using knowledge DB. Study LightGBM
Theint Su	- Preprocessed the ClaimBuster dataset by cleaning text, removing unnecessary characters, and preparing TF-IDF features - Trained Logistic Regression as a baseline model for comparison. - Trained the AdaBoost classifier using a Decision Tree base estimator. - Applied 5-fold Stratified Cross Validation and GridSearchCV for hyperparameter tuning. - Improved the AdaBoost model by increasing the decision tree depth and expanding the hyperparameter search space. Problem: Model predicts most of the texts as "label 2" [unbalanced distribution] The distribution of the features are unbalanced. [Label 0: 10.5%, Label 1: 24%, Label 2: 65.5%]

Suggestion

Minimize the part 4 - removing the random forest model and apply a simple voting system.

E.g.



Main Script Suggestion

```
def main(input: str, k: int = 5) -> None:

    check_worthy = func_part1(input)
    if not check_worthy:

        print('The claim is not check-worthy')

        return

    evidence = func_part2(input, k)
    if evidence is None or len(evidence) == 0:

        print('No evidence found')

        return

    relationships = func_part3(input, evidence)

    majority = len(evidence) // 2 + 1

    supports = [r for r in relationships if r.relation == 'SUPPORTS']
    if len(supports) >= majority:

        print('SUPPORTS')
        for support in supports:

            print(f"\tEvidence: {support.evidence}, Score: {support.score}")

        return

    refutes = [r for r in relationships if r.relation == 'REFUTES']
    if len(refutes) >= majority:

        print('REFUTES')
        for refute in refutes:

            print(f"\tEvidence: {refute.evidence}, Score: {refute.score}")

        return

    print('NOT ENOUGH EVIDENCE')

    return
```

ECE 503 Progress Report

Project Title: Fact-checking pipeline for preventing hallucinations

Members:

- | | |
|-------------------|-----------|
| 1. Robert Widjaja | V01095154 |
| 2. Chloe Lee | V01095619 |
| 3. Theint Su | V01095622 |

I. The Problem

As the usage of AI chatbots increases, many people are exposed to tons of information. According to Reuters [1], the number of people using AI to find news rises to 15% among under-25s. AI chatbots are so easy to use that everyone can access all the information without much effort. This can be interpreted as positive, because it seems like the information gap would be eliminated.

However, this has brought a serious problem, AI hallucination. The AI hallucination rate has increased about 94% from 2024 to 2025 [2], while the number of people using AI chatbots keeps increasing; the number of weekly active users of ChatGPT increased from 250M in October 2024 to 800M in October 2025 [3]. Yet, according to Statistics Canada [4], 81% of people haven't done fact-checking the information from online in the past 12 months. To mitigate the risk, AI chatbots must prevent hallucination from the system, not force users to fact-check the information by themselves.

Therefore, we decided to build a fact-checking pipeline for preventing hallucinations. This project aligns with the recent goal of many AI systems, building a RAG (Retrieval-Augmented Generation) pipeline. In addition, by building the pipeline, all team members will be able to experience the entire process of NLP and improve our knowledge.

Datasets below will be used and all of them are able to be obtained through HuggingFace. Therefore, the risk of failing to obtain the dataset will be low. However, in case of failure to obtain, we mentioned the backups. Also, additional or alternative datasets might be used during implementation.

First, [ClaimBuster](#) will be used to train the model for claim extraction. This dataset is used for evaluating if the text is check-worthy. Before analyzing the data, we considered that its labels consist of Non-factual(0), Unimportant Factual(1), and Check-worthy Factual(2). However, after further investigation of the original ClaimBuster documentation, we clarified the label definitions as Label (0) as Unimportant Factual Statement (UFS), Label (1) as Check-worthy Factual Statement (CFS) and Label (2) Non-factual Statement (NFS) or rhetorical statements. In addition, we also planned to train only the AdaBoost model as our baseline check-worthy model. However, we now try to compare between Logistic Regression and AdaBoost with cross-validation and tuned hyperparameters.

Second, to build a knowledge DB, we had planned to use [KILT Wikipedia Dump](#) for constructing the search engine. However, there was a problem when building the search engine. We tried to use AWS serverless Open Search service and load the dataset into the engine, but there were many problems when loading the data, such as resource or network issues. Therefore, we changed to using the [pyserini](#) library instead of building our own search engine.

Third, [MNLI](#) will be used to train the model for NLI and used to classify a baseline on TF-IDF + Logistic Regression. This dataset is used for finding the logical relationship between two texts. Its labels consist of Entailment(0), Neutral(1), and Contradiction(2). If there's enough time to enhance the model

for edge cases, [ANLI](#), which has the same labels as MNLI but contains tricky texts, can be used. Also, [SNLI](#) can be used as MNLI's backup dataset. Across these datasets, useless rows will be dropped and cleaned, -1 labels and empty text will be cleaned, keeping only the shared columns.

Finally, [FEVER](#) will be used to train the models for evidence retrieval and NLI. This dataset is used for classifying if the evidence supports the claim text. Its labels consist of SUPPORTS, NOT ENOUGH INFO, and REFUTES. [FEVEROUS](#) can be used as FEVER's backup dataset.

Aside from the problems with the datasets, there was a change in team members. As one of the members is not going to continue this course, we have to reduce the tasks. Therefore, we selected a simple voting system instead of building a random forest model for aggregation. Also, we will evaluate the full-pipeline performance with label accuracy instead of other measurements requiring more time and resources.

II. Goals

Our main objective is to build a functional, multi-stage fact-checking pipeline that actually works under realistic data constraints. Because each part of this system feeds directly into the next, a mistake early on will ruin everything down the line. To make sure we are on track, we are breaking our goals down into specific data benchmarks for each block, followed by clear success targets for the completed pipeline.

1. Component testing & Localised metrics

Before we link everything together, we need to make sure each stage works reliably on its own. We will measure individual component success using these specific metrics:

- **Claim Extraction:** We need to separate actual factual claims from casual banter or opinions. We will measure our model(Logistic Regression or AdaBoost) using the macro-F1 score on the ClaimBuster dataset. Our main priority is keeping the recall high for Class 1 (check-worthy text) so we do not accidentally throw out important claims before they even get a chance to be checked.
- **Evidence Retrieval:** We are scoring our LightGBM model using Recall@k and Mean Reciprocal Rank (MRR) against the KILT Wikipedia dump and DPR corpus. The goal is to tweak our ranking logic so that the exact sentence needed to verify a claim is consistently pushed into the top couple of results (targeting an MRR close to 1.0).
- **Natural Language Inference (NLI):** We will score our NLI classifier based on per-class accuracy across the three standard MNLI and ANLI categories: entailment, neutral, and contradiction. The model needs to handle tricky phrasing without taking shortcuts or defaulting to the most common class label.

2. Full-Pipeline Outcomes

For the integrated system, we will grade the final output using the label accuracy. According to A. Hanselowski [5], label accuracy only using re-rank(part 2) based on the same dataset, FEVER, is up to 68.49%. As we are building a pipeline having claim extraction, evidence retrieval, and NLI, we aim to get the final label accuracy higher than 70%.

III. Plan and Progress-to-date

Our plan will follow a 4 stage process. First, raw data is passed from test to claim, then moving onto the evidence retrieval, natural language inference before finally arriving at a final voting system. The data format to be used between each pair of trials will need to be consistent so that the four components can be independently developed and tested, and then stitched one at a time. In all, we follow the same experimental approach for each stage: we create a basic baseline to validate our data and evaluation setup, then compare that baseline with a stronger model learned using that baseline, and report on improvements alongside per-stage error analysis.

Our main strategy is to ground each stage from the foundational literature listed in the references and to reproduce its published baselines before trying to improve its performance for generalization for evaluation. We first start with using ClaimBuster [6] for check-worthiness, LightGBM [7] and the KILT benchmark [8] for evidence retrieval, Adversarial NLI [10] and Sentence-BERT [11] for natural language inference and lastly the FEVER paper [12] together with the Retrieval-Augmented Generation paper [13] for verification and overall design of our pipeline. Other than literatures, we will be reproducing the reported number as a sanity check, public leaderboards such as FEVER and MNLI as performance references, and build on well documented libraries such as scikit-learn and hugging face transformers to reduce implementation risk. We plan on using a shared github repository with a common testing suite for comparing implementations, having the data flow between modules to be stable after implementation begins. If we do have any conceptual blockers not covered in the literature, we will consult Professors who are experts in that field.

For the claim extraction, we have trained an initial AdaBoost model. However, we also want to compare with other models, such as Logistic Regression, to see if other models result in better accuracy than AdaBoost. For information retrieval, we have changed the knowledge DB from building our own search engine to importing an external library and building a proxy API server to use. The reason why we have changed is that building a knowledge DB is not directly related to Machine Learning, and there was not enough time to resolve the absurd problems when building the knowledge DB. Also, we are still studying the literature about LightGBM, and we implemented importing the dataset for the LightGBM model. For the NLI, we studied the datasets, such as MNLI, ANLI, and SNLI, and built a unified dataset. Also, we have built a baseline comparison using TF-IDF and Logistic Regression. We now started working on the cross-encoder wrapper and fine-tuning.

IV. Task Breakdown

Our project is divided into three main components. Each team member is responsible for the development, training, and evaluation of machine learning models within their assigned components. These components include Claim Extraction, Evidence Retrieval, and Natural Language Inference.

In addition to our individual responsibilities, all team members will participate in project coordination activities, including debugging, performance analysis and preparation of the final project presentation of the report. The detailed individual tasks are as follows:

- (1) Theint is responsible for Claim Extraction. This task includes preprocessing data for claim extraction, developing an AdaBoost model, training and fine-tuning the model and evaluating claim extraction performance.
- (2) Chloe is responsible for Evidence Retrieval and Aggregation. This task includes importing the search engine, preprocessing data for evidence retrieval, studying and implementing a LightGBM model, fine-tuning and evaluating the retrieval model, and implementing the main script.
- (3) Robert is responsible for Natural Language Inference. This task includes studying the MNLI Dataset and the Sentence-BERT architecture, performing data preprocessing, developing the NLI model using cross encoders, training, fine-tuning the model and evaluating natural language inference performance.

V. Initial results

(1) Baseline results for NLI for MNLI

To validate the data pipeline and evaluation setup before spending compute on a transformer, I first implemented a TF-IDF + Logistic Regression baseline. The baseline uses word 1–2 grams over the concatenation premise [SEP] hypothesis, augmented with two hand-crafted features (hypothesis-token overlap ratio with the premise, and length delta). It was trained on 100,000 examples sampled from MNLI's training split and evaluated on the full 9,815-example MNLI-matched development set.

Metric	Value
Accuracy	0.5205
Macro-F1	0.5195
F1 entailment	0.5397
F1 neutral	0.5246
F1 contradiction	0.4943

The overall accuracy (52%) sits well above the 33% random-guessing floor for a balanced 3-way classification task, confirming that the data and evaluation loop are wired correctly. More importantly, all three per-class F1 scores are within roughly four points of each other, the model is not defaulting to the most frequent class, which is the specific shortcut behavior our proposal warned about ("The model needs to handle tricky phrasing without taking shortcuts or defaulting to the most common class label"). This makes it a meaningful floor for the neural cross-encoder to beat.

(2) Baseline results (NLI - ANLI)

	precision	recall	F1-score	support
entailment	0.2347	0.2066	0.2197	334
neutral	0.3354	0.4865	0.3971	333
contradiction	0.2601	0.1742	0.2086	333

As expected, the results for the ANLI dataset for the TF-IDF + Logistic Regression baseline is far worse than using the MNLI due the nature of ANLI containing tricky sentences. Using these baseline results, we can compare the fine tuned cross-encoder (work in progress) with Sentence-BERT if it is better than simple “dumb” baseline algorithms.

(3) Baseline Results for ClaimBuster Dataset)

Experiment	Dataset	Accuracy	Precision	Recall	F1 score
Logistic Regression	Original	0.7699	0.6852	0.5698	0.5972
AdaBoost	Original	0.7253	0.6341	0.4787	0.4978

The current result shows that Logistic Regression still has higher accuracy than AdaBoost Model after tuning hyperparameters.

An additional issue observed during evaluation is that both Logistic Regression and AdaBoost models cannot classify Label 0 correctly and misclassified as Label 2. One possible reason for this behaviour is the class imbalance in the ClaimBuster Dataset where Label 2 contains more samples than the other classes. Therefore, we will use Random Sampling and train the model again with a balanced dataset to check if this improves the classification performance.

VI. References

- [1] David Elliott, “This is how people in 2025 are getting their news,” *World Economic Forum*, Jul. 14, 2025. <https://www.weforum.org/stories/2025/07/news-consumption-social-video/>
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- [13] P. Lewis *et al.*, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *NeurIPS*, 2020. <https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html>